Arunabha Majumder

Mechanism Design Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University | 2023 - present
Thesis expected March

MTech, Mechatronics
AcSIR - CSIR-CMERI, India
2020 - 2022

BTech, Mechanical Engineering
Siddaganga Institute of Technology, India
2015 - 2019

Senior Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad
2008, First Division with Distinction

STUDY ABROAD

National University of Singapore
Sep - Nov 2025 (during my PhD)

SKILLS

CAD: SolidWorks (CSWA), technical drawings, stiffness analysis in SolidWorks, Fusion 360 (basic, last used 2019)

Mechanisms: kinematic, torque and force calculations, Jacobian-based force estimation, worm gears, springs, GA optimisation

Build and test: 3D printing, assembly, test setups, motors, encoders, force/torque sensors, Arduino, MATLAB/Simulink, LabVIEW, Qualisys

Art and design: hand sketching, painting, Adobe Illustrator, Photoshop, Procreate

ACHIEVEMENTS

Best Paper Award, IFToMM ISRM 2026

Best Major Project Award, 2019

TATA Technologies Prayatna grant

GATE qualified

LANGUAGES

English (C1), Bengali (native), Hindi, German (willing to learn)

REFERENCES

Prof. Shaoping Bai
Aalborg University, sbh@mp.aau.dk

Assoc. Prof. Lars Rosgaard Jensen
Head of Section and Head of Labs, AAU

Assoc. Prof. Anderson de Souza Castelo Oliveira
Materials and Production, AAU

PROFILE

I am a mechanical engineer who designs mechanisms and then tests them on real hardware. Over the last 6 years I have worked on variable stiffness mechanisms, a worm gear pretension system, a 5-bar parallel robot and exoskeletons, from the first hand calculations and sketches to CAD, prototypes and test results. I enjoy the kinematic, torque and force side of the work, writing it up clearly, and I would love to bring this to mechanisms that fly.

EXPERIENCE

Research Assistant

Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton: a variable stiffness mechanism (spring pretensioned by a worm gear) working in parallel with a 6 Nm motor; together they give the torque needed to lift the arm. It uses over 25% less motor energy (Best Paper Award, IFToMM ISRM 2026).
- Optimised the mechanism with a genetic algorithm (diameter, spring stiffness, pretension force), made the CAD and drawings in SolidWorks, 3D printed the first parts and had the final parts manufactured.
- Integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical setup.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance. Force estimation used the variable stiffness actuators and the robot's Jacobian.
- Arm impedance study with the 5-bar robot (6 participants): Qualisys cameras tracked the arm, and I implemented an Arduino-based trigger to sync the robot's motion with the recording.
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit

CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit. Designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Bachelor's Projects

Siddaganga Institute of Technology, India | *2017 - 2019*

- Built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly and testing.
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

PUBLICATIONS

- Hybrid actuated shoulder exoskeleton, IFToMM ISRM 2026 (Springer), Best Paper Award.
- Neural network-based control of a pneumatic artificial muscle, IEEE CONECCT 2022.

PORTFOLIO

Videos and results from my work: arunabha22.github.io/arunabha-majumder